
4. Troubleshooting

4-1. Checkpoints by Error Mode

Oscilloscope Setting Values	Check DC voltage	Check I ² C& I ² S signal	check PWM signal	Check L+,L- signal
Voltage/DIV	Depend voltage	1V/div	10V/div	10V/div
TIME/DIV	500ms/div	20us/div	1us/div	500us/div

4-1-1. No Power

Symptom	• If you connect the power cables, the LED on the front will turn on • If the units do not work
Major checkpoints	If 18V not working
Diagnostics	Flowchart for No Power diagnosis: <pre>graph TD; A{Check the power board output CN2 voltage is 18V?} -- YES --> B{Check whether replace FUSE and Varistor is ok?}; A -- NO --> D{Replace power board}; B -- YES --> C{Replace the FUSE and Varistor}; B -- NO --> D;</pre> Image of the power board with labels for Varistor, FUSE, and CN2. Circuit diagram of the power board output CN2, showing a 200uH inductor (LF2) and a 0.1uF 150V X7R capacitor (C10) connected to the CN2 connector pins 1, 2, 3, and 4.
Caution	• Before working, disconnect the power cord.

4-1-1. No Power

Symptom	- If you connect the power cables, the LED on the front will turn on - If the units do not work
Major checkpoints	If STB_5V not working
Diagnostics	NO STB_5V YES Check the power board output voltage is 18V— check voltage pin 1or2 of XP1? NO replace power board YES Replace U1
Caution	Before working, disconnect the power cord.

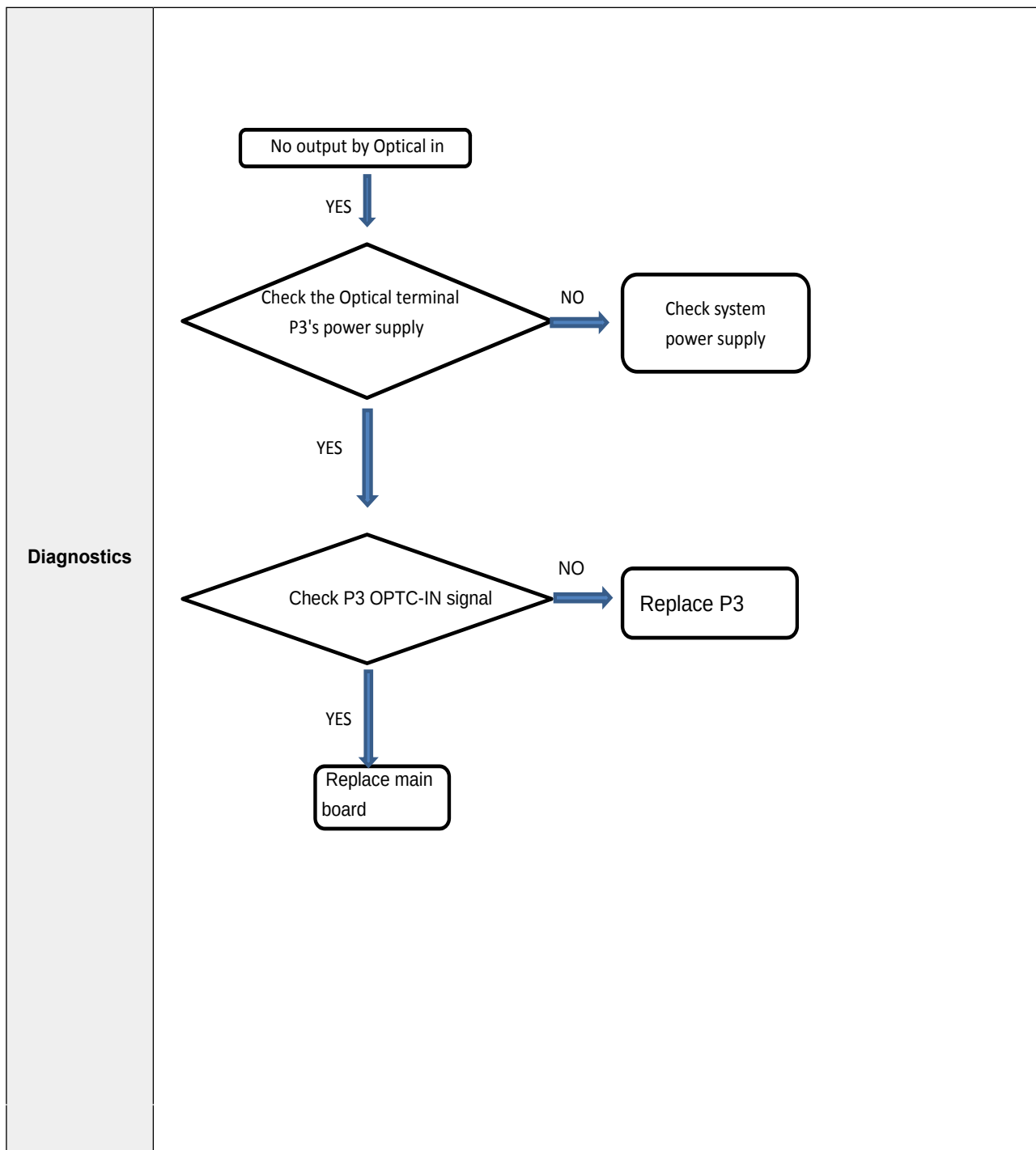
4-1-1. No Power

Symptom	- If you connect the power cables, the LED on the front will turn on - If the units do not work
Major checkpoints	If STB_5V working properly
Diagnostics	<pre> graph TD Start([NO STB_3.3V]) -- YES --> D1{Is STB_5V supplied to pin1, pin4 of U3?} D1 -- NO --> C1[Check whether R14 is open circuit or C14, C15 is short circuit] D1 -- YES --> D2{Is U3's pin3 short circuit to ground?} D2 -- NO --> C2[Check whether L2 is open circuit or C12, C13 short circuit] D2 -- YES --> End([Replace U3]) </pre>
Caution	Before working, disconnect the power cord.

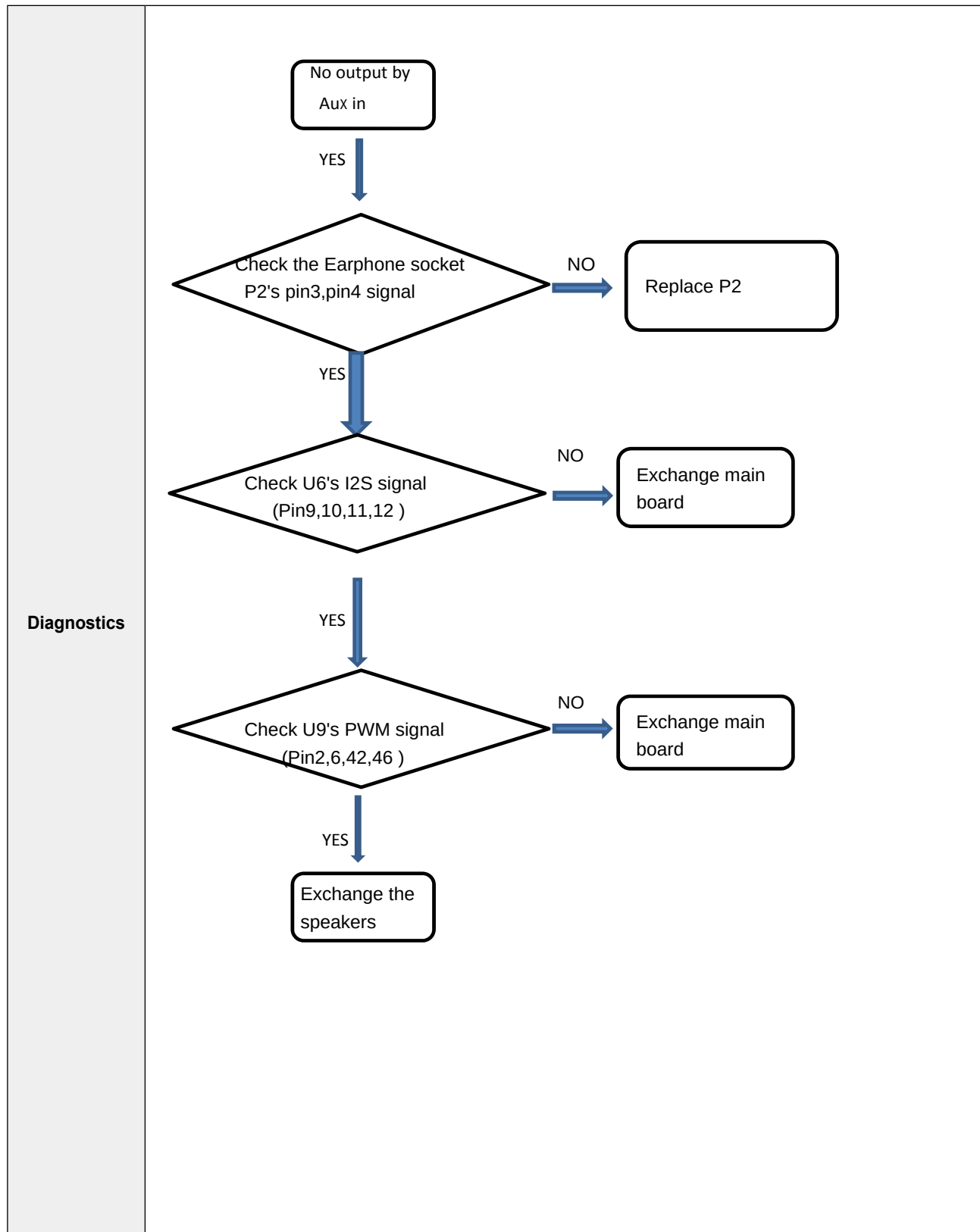
4-1-2. No Audio Output

Symptom	If the sound is not output.
Major checkpoints	- Speaker connector is not connected properly or check for damage. - Make sure your sound processing functions operate properly on the motherboard. - Make sure the speaker isn't bad.
Diagnostics	<pre> graph TD Start([NO audio output]) -- YES --> D1{Is 18V on Main board is normal?} D1 -- NO --> R1[Replace power board] D1 -- YES --> D2{Are the signals of I2S (U6 pin 9,10,11,12) normal?} D2 -- NO --> R2[Refer to No 3.3V detecting steps] D2 -- YES --> D3{Are the PWM signals (U9 pin 4,6,43,46) normal?} D3 -- NO --> R3[Exchange main board] D3 -- YES --> R4[Replace the Speakers] </pre> The flowchart starts with a box labeled "NO audio output". It then proceeds to a decision diamond: "Is 18V on Main board is normal?". If the answer is "NO", the action is "Replace power board". If "YES", it goes to the next decision diamond: "Are the signals of I2S (U6 pin 9,10,11,12) normal?". If "NO", the action is "Refer to No 3.3V detecting steps". If "YES", it goes to the third decision diamond: "Are the PWM signals (U9 pin 4,6,43,46) normal?". If "NO", the action is "Exchange main board". If "YES", the final action is "Replace the Speakers".
Caution	Before working, disconnect the power cord.

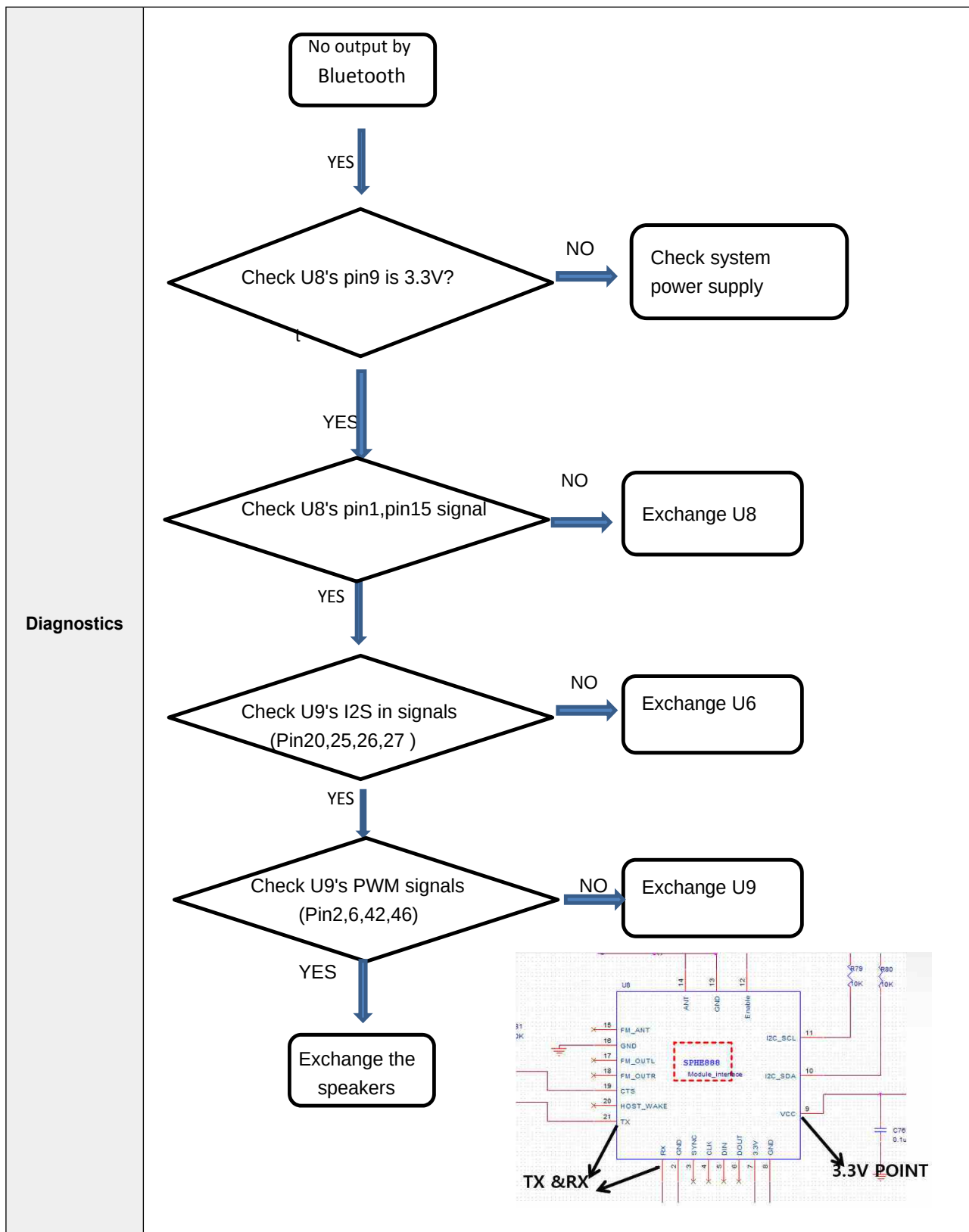
4-1-3. No output by Optical in



4-1-4. No Sound Output by Aux in



4-1-5. In case of No output by BT



4-1-6. LED Module Failure Analysis

Start

YES

Check if 3.3V power is measured at pin 7,15,16 of XP18 on the Main PCB.

NO

Replace XP18 & FFC Cable Check

NO

Change MAIN PCB

YES

Check if signal is measured at pin No. 1,2,3,4,5,6,8,10 of XP18 on the Main PCB.

NO

Replace XP18 & FFC Cable Check

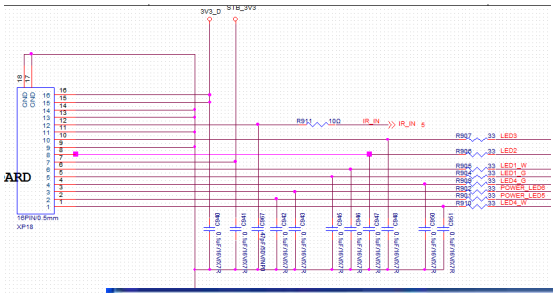
NO

Change MAIN PCB

YES

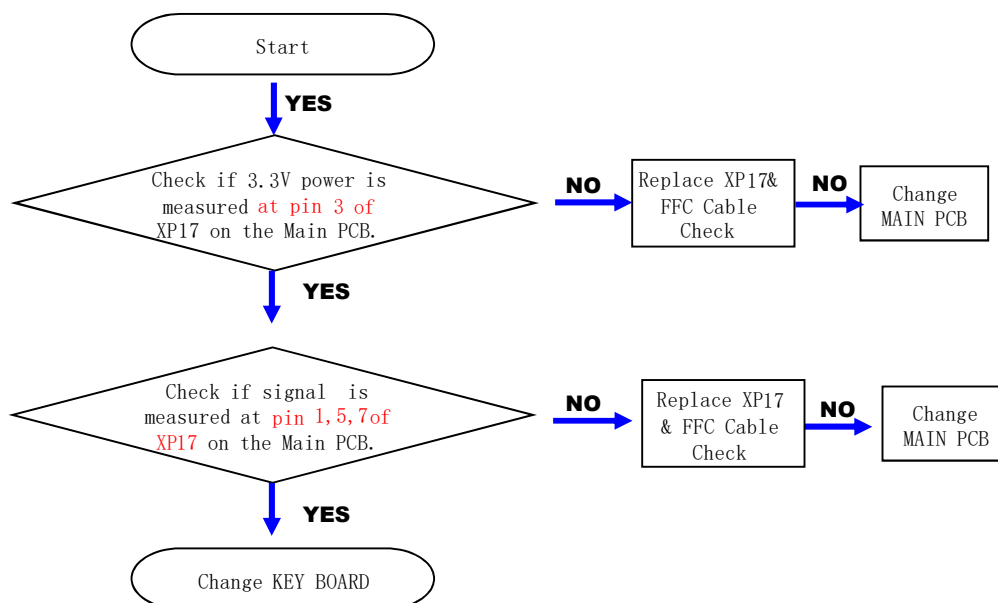
Change LED BOARD

Diagnostics

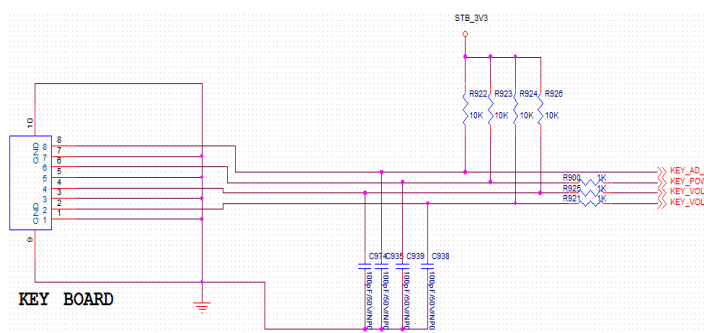


XP19 , to MB	
1	3.3V_D
2	3.3V_D
3	GND
4	GND
5	IR_IN
6	GND
7	LED3
8	GND
9	LED2
10	STB_3.3V
11	LED1_W
12	LED1_G
13	LED4_G
14	POWER_LED6
15	POWER_LED5
16	LED4_W

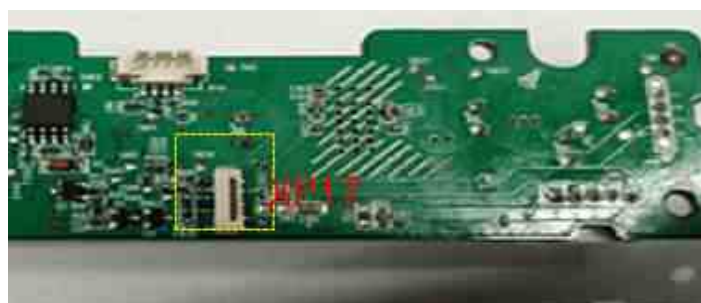
4-1-7. KEY Module Failure Analysis



Diagnostics

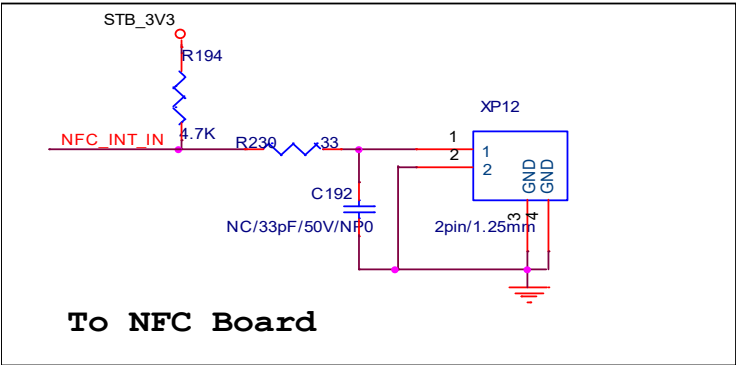
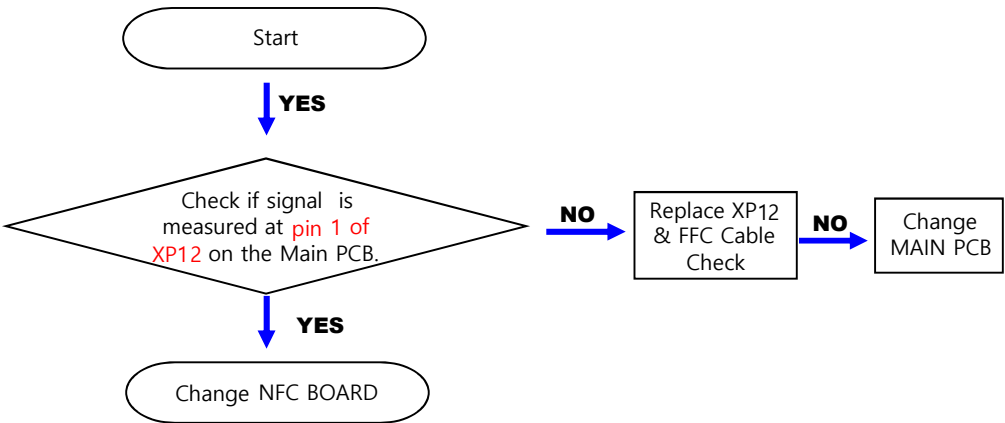


XP17 to KEY	
1	KEY_AD_IN
2	GND
3	KEY_POWER
4	GND
5	KEY_VOL+
6	GND
7	KEY_VOL-
8	GND

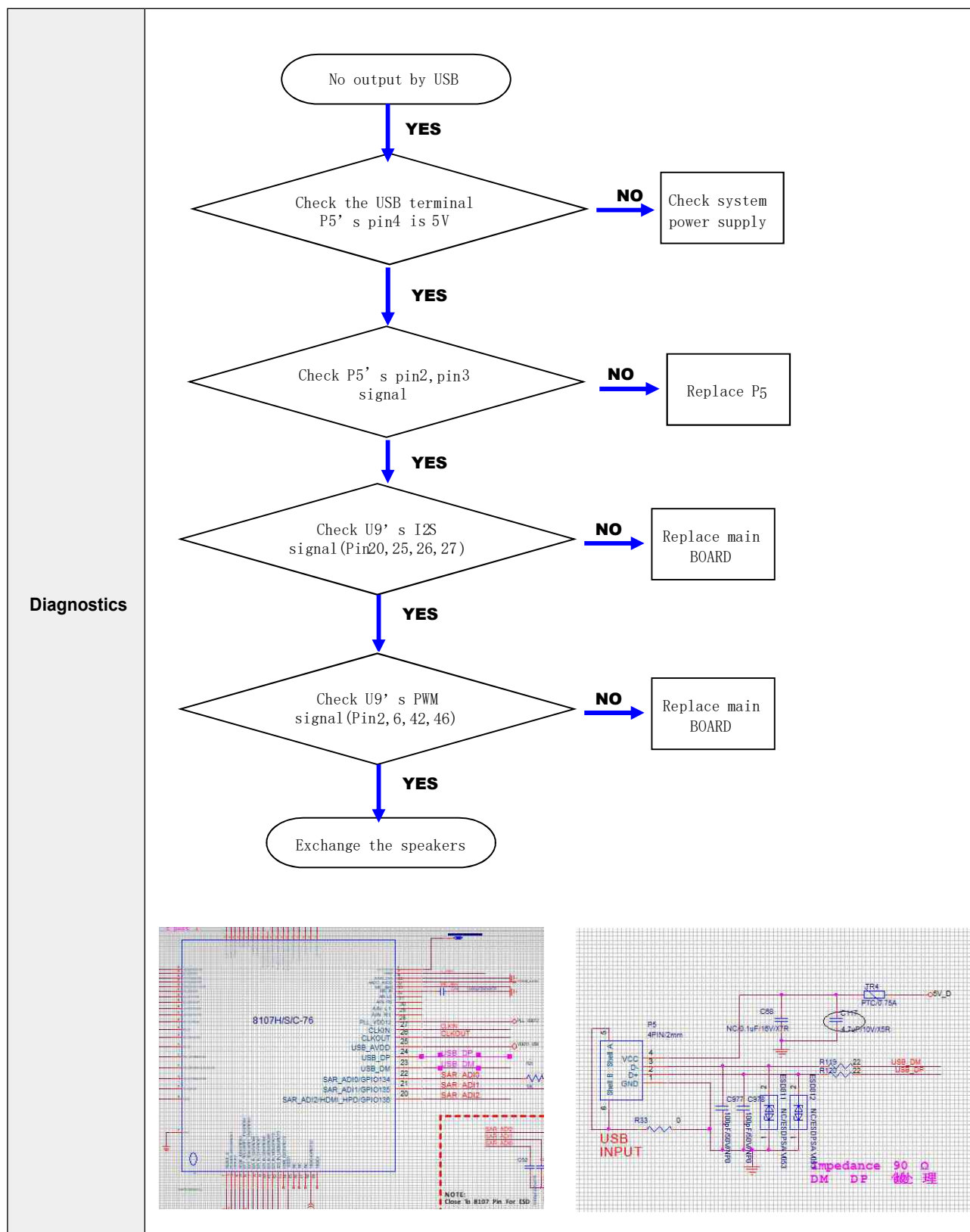


4-1-8. NFC Module Failure Analysis

Diagnostics



4-1-9. In case of No output by USB



4-2. Measures to be taken when the Protection Circuit operates

4-2-1. AMP Pre-Inspection relating to Power Protection

If you think there are problems at the AMP PCB, you can check the PCB without disassembling the set by following the test below.

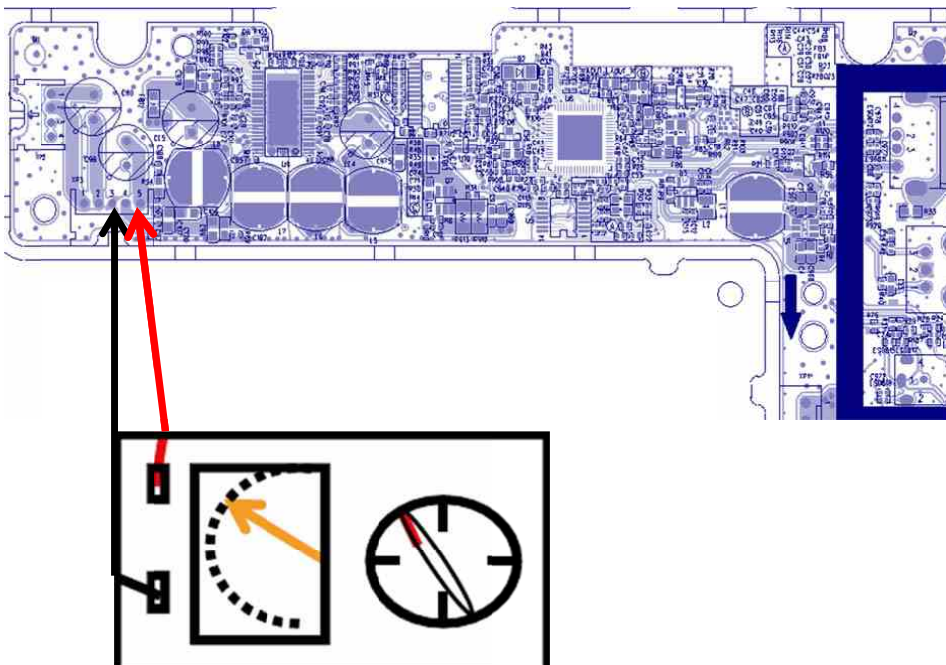


Caution

- Do not connect the power cord during the test!

Resistance using Tester	
F/R CH	9.48M ohm

- If there is a large difference than the value listed above then the AMP PCB has a problem.



4-3. Initialization & Update

4-3-1. Set Initialization method

1. Sound Bar(Using the remote control)

- 1.Prepare the remote control.
2. Turn the unit off.
3. Press and hold the “Play/Pause” button on the remote for more than 5 seconds.
4. If it was successful, the whole led will be blink twice and turned off.

2. Sound Bar (Without Using the Remote control)

- 1.Turn the unit on.
2. Press and hold the “+” button and “-” button on Sound Bar simultaneously for 5 seconds.
3. If it was successful, the whole led will be blink twice.
4. The Sound bar will be reboot up.

4-3. Initialization & Update

4-3-2. How to check firmware version

- 1. Prepare the remote control.
- 2. Turn the unit off.
- 3. Press the “USB” button for 5 seconds in the remote control.
- 4.The unit will power on and show LED light sequence.
- 5.See table and example below.
- 6.The unit will return to current source mode after step5.

The effect of running water lamp is as below:

Power On Ceremony	LED1	LED2	LED3	LED4	LED5
0-0.5s					
0.5-1.0s					
1.0-1.5s					
1.5-2.0s					
2.0-2.5s					

The quinary method:The LED1 is as 5^4 . The LED2 is as 5^3 . The LED3 is as 5^2 . The LED4 is as 5^1 . The LED5 is as 5^0 .

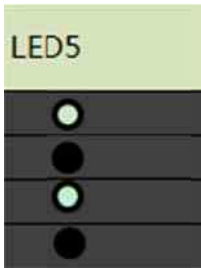
For example the version 12, LED4 will blink twice, LED5 will blink twice, too.

Version 12 = $5^1 * 2 + 5^0 * 2$.

The LED4 is shown as below.



After LED4 is completed, LED5 is shown as below.



4-3-3. USB Update procedure

1 Prepare USB upgrade file

- 1.1 Use an empty USB flash drive.
- 1.2 Copy the upgrade bin to the root directory of USB flash drive.
- 1.3 Confirm the file name is "HMS-N300WWB-XXX.bin". Where the "XXX" refers to the version number.

2 Power on the HW-N300 sound bar.

Plug in power, and then press the  button on the remote control or press the  button on the front-panel of soundbar.

3 Enter upgrade mode.

Short press the USB button of remote control to enter the USB source. The led display is as below.



4 Plug in the USB flash drive and start upgrade.

Plug in the USB flash drive and wait for about 8 seconds. If the sound bar find the correct upgrade file it will be upgrade automatically. During the upgrade process the led display is as below.



During the upgrade process don't power down the HW-N300 sound bar and wait for it upgrade successfully.

5 Finish and reboot up.

Once upgrade successfully, sound bar will be reboot up directly. The USB source led is solid after the reboot. The led display is as step 3.

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